

ETC Reference Guide

High End Systems Lonestar DMX Channel Map

Lonestar DMX Channel Map (Software Version 1.6.3 and Later)

Standard Protocol					
Channel	Function	Channel	Function	Channel	Function
1	Pan Coarse	17	Blade 1 Angle A	33	Prism 2 Rotate Coarse
2	Pan Fine	18	Blade 1 Angle B	34	Prism 2 Rotate Fine
3	Tilt Coarse	19	Blade 2 Angle A	35	Diffusion Light
4	Tilt Fine	20	Blade 2 Angle B	36	Diffusion Medium
5	Color Mode	21	Blade 3 Angle A	37	Focus Coarse
6	Cyan	22	Blade 3 Angle B	38	Focus Fine
7	Magenta	23	Blade 4 Angle A	39	Zoom Coarse
8	Yellow	24	Blade 4 Angle B	40	Zoom Fine
9	CTO	25	Frame Rotation Coarse	41	Iris
10	Color Wheel Mode	26	Frame Rotation Fine	42	Strobe Mode
11	Color Wheel	27	Animation Insertion	43	Strobe
12	Rotating Gobo Wheel Mode	28	Animation	44	Dimmer Coarse
13	Rotating Gobo Wheel	29	Prism 1 Mode	45	Dimmer Fine
14	Rotating Gobo Mode	30	Prism 1 Rotate Coarse	46	mSpeed
15	Rotating Gobo Coarse	31	Prism 1 Rotate Fine	47	Control
16	Rotating Gobo Fine	32	Prism 2 Mode	48	Fan

Trifusion Frost Protocol					
Channel	Function	Channel	Function	Channel	Function
1	Pan Coarse	17	Blade 1 Angle A	33	Diffusion Heavy Coarse
2	Pan Fine	18	Blade 1 Angle B	34	Diffusion Heavy Fine
3	Tilt Coarse	19	Blade 2 Angle A	35	Diffusion Light
4	Tilt Fine	20	Blade 2 Angle B	36	Diffusion Medium
5	Color Mode	21	Blade 3 Angle A	37	Focus Coarse
6	Cyan	22	Blade 3 Angle B	38	Focus Fine
7	Magenta	23	Blade 4 Angle A	39	Zoom Coarse
8	Yellow	24	Blade 4 Angle B	40	Zoom Fine
9	CTO	25	Frame Rotation Coarse	41	Iris
10	Color Wheel Mode	26	Frame Rotation Fine	42	Strobe Mode
11	Color Wheel	27	Animation Insertion	43	Strobe
12	Rotating Gobo Wheel Mode	28	Animation	44	Dimmer Coarse
13	Rotating Gobo Wheel	29	Prism 1 Mode	45	Dimmer Fine
14	Rotating Gobo Mode	30	Prism 1 Rotate Coarse	46	mSpeed
15	Rotating Gobo Coarse	31	Prism 1 Rotate Fine	47	Control
16	Rotating Gobo Fine	32	Reserved	48	Fan



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Lonestar Standard Protocol

Channel	Function	Description	Decimal Low	Decimal High	Percent Low	Percent High	Hex Low	Hex High	Controller Default
1/2	Pan (Coarse/Fine)	See note 7 on page 16	0	65535	0%	100%	0000h	FFFFh	32767
3/4	Tilt (Coarse/Fine)	See note 7 on page 16	0	65535	0%	100%	0000h	FFFFh	32767
5	Color Mode	Full Speed Control							0
		Pure Mix	0	31	0%	12%	00h	1Fh	
		Cycle	32	47	13%	18%	20h	2Fh	
		Random	48	63	19%	25%	30h	3Fh	
		Pure Mix Fast	64	79	25%	31%	40h	4Fh	
		Cycle Fast	80	95	31%	37%	50h	5Fh	
		Random Fast	96	111	38%	44%	60h	6Fh	
		Reserved (see note 5 on page 15)	112	127	44%	50%	70h	7Fh	
		mSpeed Control							
		Pure Mix	128	159	50%	62%	80h	9Fh	
		Cycle	160	175	63%	69%	A0h	AFh	
		Random	176	191	69%	75%	B0h	BFh	
		Pure Mix Fast	192	207	75%	81%	C0h	CFh	
		Cycle Fast	208	223	82%	87%	D0h	DFh	
		Random Fast	224	239	88%	94%	E0h	EFh	
		Reserved (see note 5 on page 15)	240	255	94%	100%	F0h	FFh	
6 7 8	Cyan Magenta Yellow	Pure Mix							255
		Full Saturation to Open	0	255	0%	100%	00h	FFh	
		Cycle and Random. Scan speed controlled by Cyan channel.							
9	CTO	Slow to Fast	0	255	0%	100%	00h	FFh	255
		Full Saturation to Open	0	255	0%	100%	00h	FFh	

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Channel	Function	Description	Decimal Low	Decimal High	Percent Low	Percent High	Hex Low	Hex High	Controller Default	
10	Color Wheel Mode	Full Speed Control								48
		Indexed (see note 1 on page 15)	0	15	0%	6%	00h	0Fh		
		Forward Spin	16	31	6%	12%	10h	1Fh		
		Reverse Spin	32	47	13%	18%	20h	2Fh		
		Continuous (see note 1 on page 15)	48	63	19%	25%	30h	3Fh		
		Scan	64	79	25%	31%	40h	4Fh		
		Random	80	95	31%	37%	50h	5Fh		
		Reserved (see note 5 on page 15)	96	127	38%	50%	60h	7Fh		
		mSpeed Control								
		Indexed (see note 1 on page 15)	128	143	50%	56%	80h	8Fh		
		Forward Spin	144	159	56%	62%	90h	9Fh		
		Reverse Spin	160	175	63%	69%	A0h	AFh		
		Continuous (see note 1 on page 15)	176	191	69%	75%	B0h	BFh		
		Scan	192	207	75%	81%	C0h	CFh		
		Random	208	239	82%	94%	D0h	EFh		
		Reserved (see note 5 on page 15)	240	255	94%	100%	F0h	FFh		

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Channel	Function	Description	Decimal Low	Decimal High	Percent Low	Percent High	Hex Low	Hex High	Controller Default
11	Color Wheel	Indexed							0
		Open	0	10	0%	4%	00h	0Ah	
		Open/Red	11	21	4%	8%	0Bh	15h	
		Red	22	32	9%	13%	16h	20h	
		Red/Blue	33	43	13%	17%	21h	2Bh	
		Blue	44	54	17%	21%	2Ch	36h	
		Blue/Green	55	65	22%	25%	37h	41h	
		Green	66	76	26%	30%	42h	4Ch	
		Green/Yellow	77	87	30%	34%	4Dh	57h	
		Yellow	88	98	35%	38%	58h	62h	
		Yellow/Purple	99	109	39%	43%	63h	6Dh	
		Purple	110	120	43%	47%	6Eh	78h	
		Purple/TM-30	121	131	47%	51%	79h	83h	
		TM-30	132	142	52%	56%	84h	8Eh	
		TM-30/Lavender	143	153	56%	60%	8Fh	99h	
		Lavender	154	164	60%	64%	9Ah	A4h	
		Lavender/Half CTO	165	175	65%	69%	A5h	AFh	
		Half CTO	176	186	69%	73%	B0h	BAh	
		Half CTO/Full CTO	187	197	73%	77%	BBh	C5h	
		Full CTO	198	208	78%	82%	C6h	D0h	
		Full CTO/Dark Blue	209	219	82%	86%	D1h	DBh	
		Dark Blue	220	230	86%	90%	DCh	E6h	
		Dark Blue/Open	231	241	91%	95%	E7h	F1h	
		Open	242	255	95%	100%	F2h	FFh	
		Scan							
		Open/Red, Slow to Fast	0	21	0%	8%	00h	15h	
		Red/Blue, Slow to Fast	22	43	9%	17%	16h	2Bh	
		Blue/Green, Slow to Fast	44	65	17%	25%	2Ch	41h	
		Green/Yellow, Slow to Fast	66	87	26%	34%	42h	57h	
		Yellow/Purple, Slow to Fast	88	109	35%	43%	58h	6Dh	
		Purple/TM-30, Slow to Fast	110	131	43%	51%	6Eh	83h	
		TM-30/Lavender, Slow to Fast	132	153	52%	60%	84h	99h	
		Lavender/Half CTO, Slow to Fast	154	175	60%	69%	9Ah	AFh	
		Half CTO/Full CTO, Slow to Fast	176	197	69%	77%	B0h	C5h	
		Full CTO/Dark Blue, Slow to Fast	198	219	78%	86%	C6h	DBh	
		Dark Blue/Open, Slow to Fast	220	255	86%	100%	DCh	FFh	
		Spin and Random							
		Stop	0	0	0%	0%	00h	00h	
		Slow to Fast	1	255	0%	100%	01h	FFh	
		Continuous							
		Positioning from 0° to 359°	0	255	0%	100%	00h	FFh	

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Channel	Function	Description	Decimal Low	Decimal High	Percent Low	Percent High	Hex Low	Hex High	Controller Default	
12	Rotating Gobo Wheel Mode	Full Speed Control								0
		Indexed	0	15	0%	6%	00h	0Fh		
		Forward Spin	16	31	6%	12%	10h	1Fh		
		Reverse Spin	32	47	13%	18%	20h	2Fh		
		Scan	48	63	19%	25%	30h	3Fh		
		Random	64	79	25%	31%	40h	4Fh		
		Reserved (see note 5 on page 15)	80	127	31%	50%	50h	7Fh		
		mSpeed Control								
		Indexed	128	143	50%	56%	80h	8Fh		
		Forward Spin	144	159	56%	62%	90h	9Fh		
		Reverse Spin	160	175	63%	69%	A0h	AFh		
		Scan	176	191	69%	75%	B0h	BFh		
		Random	192	207	75%	81%	C0h	CFh		
		Reserved (see note 5 on page 15)	208	255	82%	100%	D0h	FFh		
13	Rotating Gobo Wheel (see list of gobos on page 17)	Indexed and Scan								0
		Open	0	15	0%	6%	00h	0Fh		
		Gobo 1 - Bar	16	31	6%	12%	10h	1Fh		
		Gobo 2 - Starvolver	32	47	13%	18%	20h	2Fh		
		Gobo 3 - Fracked	48	63	19%	25%	30h	3Fh		
		Gobo 4 - Broken Tunnel	64	79	25%	31%	40h	4Fh		
		Gobo 5 - Seashell	80	95	31%	37%	50h	5Fh		
		Gobo 6 - Glacial	96	111	38%	44%	60h	6Fh		
		Gobo 7 - Shower	112	127	44%	50%	70h	7Fh		
		Gobo 8 - Ice	128	143	50%	56%	80h	8Fh		
		Gobo 9 - Branch Out	144	159	56%	62%	90h	9Fh		
		Reserved (see note 5 on page 15)	160	255	63%	100%	A0h	FFh		
		Spin and Random								
		Stop	0	3	0%	1%	00h	03h		
		Slow to Fast	4	255	2%	100%	04h	FFh		
		14	Rotating Gobo Mode	Indexed	0	15	0%	6%	00h	
Forward Spin	16			31	6%	12%	10h	1Fh		
Reverse Spin	32			47	13%	18%	20h	2Fh		
Forward Animate	48			63	19%	25%	30h	3Fh		
Reverse Animate	64			79	25%	31%	40h	4Fh		
Reserved (see note 5 on page 15)	80			255	31%	100%	50h	FFh		
15/16	Rotating Gobo (Coarse/Fine)	Indexed								32767
		0° to 359°	0	65535	0%	100%	0000h	FFFFh		
		Spin and Animate								32767
		Stop	0	2047	0%	3%	0000h	07FFh		
Slow to Fast	2048	65535	3%	100%	0800h	FFFFh				
17	Blade 1 Angle A (see note 6 on page 15)	Removed to Inserted	0	255	0%	100%	00h	FFh	0	

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Channel	Function	Description	Decimal Low	Decimal High	Percent Low	Percent High	Hex Low	Hex High	Controller Default
18	Blade 1 Angle B (see note 6 on page 15)	Removed to Inserted	0	255	0%	100%	00h	FFh	0
19	Blade 2 Angle A (see note 6 on page 15)	Removed to Inserted	0	255	0%	100%	00h	FFh	0
20	Blade 2 Angle B (see note 6 on page 15)	Removed to Inserted	0	255	0%	100%	00h	FFh	0
21	Blade 3 Angle A (see note 6 on page 15)	Removed to Inserted	0	255	0%	100%	00h	FFh	0
22	Blade 3 Angle B (see note 6 on page 15)	Removed to Inserted	0	255	0%	100%	00h	FFh	0
23	Blade 4 Angle A (see note 6 on page 15)	Removed to Inserted	0	255	0%	100%	00h	FFh	0
24	Blade 4 Angle B (see note 6 on page 15)	Removed to Inserted	0	255	0%	100%	00h	FFh	0
25/26	Frame Rotation (Coarse/Fine)	-60° to 60°	0	65535	0%	100%	00h	FFFFh	32767
27	Animation Insertion	Removed to Inserted	0	255	0%	100%	00h	FFh	0
28	Animation	Forward Spin, Stop to Fast	0	63	0%	25%	00h	3Fh	0
		Reverse Spin, Stop to Fast	64	127	25%	50%	40h	7Fh	
		Forward Animate, Stop to Fast	128	191	50%	75%	80h	BFh	
		Reverse Animate, Stop to Fast	192	255	75%	100%	C0h	FFh	
29	Prism 1 Mode (see note 4 on page 15)	Removed	0	15	0%	6%	00h	0Fh	0
		Continuous	16	31	6%	12%	10h	1Fh	
		Forward Spin	32	47	13%	18%	20h	2Fh	
		Reverse Spin	48	63	19%	25%	30h	3Fh	
		Forward Animate	64	79	25%	31%	40h	4Fh	
		Reverse Animate	80	95	31%	37%	50h	5Fh	
		Reserved (see note 5 on page 15)	96	255	38%	100%	60h	FFh	
30/31	Prism 1 Rotate (Coarse/Fine)	Continuous							32767
		0° to 359°	0	65535	0%	100%	0000h	FFFFh	
		Spin and Animate							0
		Stop	0	1027	0%	2%	0000h	0403h	
		Slow to Fast	1028	65535	2%	100%	0404h	FFFFh	
32	Prism 2 Mode (see note 4 on page 15)	Removed	0	15	0%	6%	00h	0Fh	0
		Continuous	16	31	6%	12%	10h	1Fh	
		Forward Spin	32	47	13%	18%	20h	2Fh	
		Reverse Spin	48	63	19%	25%	30h	3Fh	
		Forward Animate	64	79	25%	31%	40h	4Fh	
		Reverse Animate	80	95	31%	37%	50h	5Fh	
		Reserved (see note 5 on page 15)	96	255	38%	100%	60h	FFh	

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Channel	Function	Description	Decimal Low	Decimal High	Percent Low	Percent High	Hex Low	Hex High	Controller Default	
33/34	Prism 2 Rotate (Coarse/Fine)	Continuous							32767	
		0° to 359°	0	65535	0%	100%	0000h	FFFFh		
		Spin and Animate							0	
		Stop	0	1027	0%	2%	0000h	0403h		
		Slow to Fast	1028	65535	2%	100%	0404h	FFFFh		
35	Diffusion Light	Removed	0	0	0%	0%	00h	00h	0	
		Hard to Soft Edge	1	255	0%	100%	01h	FFh		
36	Diffusion Medium	Removed	0	0	0%	0%	00h	00h	0	
		Hard to Soft Edge	1	255	0%	100%	01h	FFh		
37/38	Focus (Coarse/Fine)	Focus	0	65535	0%	100%	00h	FFFFh	0	
39/40	Zoom (Coarse/Fine)	3.8° to 55°	0	65535	0%	100%	00h	FFFFh	0	
41	Iris	Closed to Open	0	255	0%	100%	00h	FFh	255	
42	Strobe Mode	Normal	0	31	0%	12%	00h	1Fh	0	
		Random	32	63	13%	25%	20h	3Fh		
		Synchronous Random (see note 2 on page 15)	64	95	25%	37%	40h	5Fh		
		Reserved (see note 5 on page 15)	96	255	38%	100%	60h	FFh		
43	Strobe	Closed	0	23	0%	9%	00h	17h	255	
		Slow to Fast	24	229	9%	90%	18h	E5h		
		Open	230	255	90%	100%	E6h	FFh		
44/45	Dimmer (Coarse/Fine)	Off to Full	0	65535	0%	100%	00h	FFFFh	0	
46	mSpeed	Not Enabled	0	3	0%	1%	00h	03h	0	
		Longest to Shortest	4	255	2%	100%	04h	FFh		
47	Control	No hold time requirement for the following functions (see note 8 on page 16):								0
		Idle	0	15	0%	6%	00h	0Fh		
		Pan and Tilt mSpeed Off	16	19	6%	7%	10h	13h		
		Reserved (see note 5 on page 15)	20	31	8%	12%	14h	1Fh		
		Hold time requirement for the following functions:								
		Display Off (hold 3 seconds)	32	47	13%	18%	20h	2Fh		
		Display On (hold 3 seconds)	48	63	19%	25%	30h	3Fh		
		Reserved (see note 5 on page 15)	64	79	25%	31%	40h	4Fh		
		Home All (hold 3 seconds)	80	95	31%	37%	50h	5Fh		
		Shutdown (hold 9 seconds)	96	111	38%	44%	60h	6Fh		
		Disable Pan/Tilt Motors (hold 3 seconds)	112	127	44%	50%	70h	7Fh		
		Dimming Mode - 2.4 kHz (hold 3 seconds)	128	143	50%	56%	80h	8Fh		
		Dimming Mode - 16 kHz (hold 3 seconds)	144	159	56%	62%	90h	9Fh		
		Reserved (see note 5 on page 15)	160	255	63%	100%	A0h	FFh		

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Channel	Function	Description	Decimal Low	Decimal High	Percent Low	Percent High	Hex Low	Hex High	Controller Default
48	Fan (see note 3 on page 15)	Idle	0	15	0%	6%	00h	0Fh	0
		Set Continuous Speed - Slow to Fast	16	207	6%	81%	10h	CFh	
		Auto	208	223	82%	87%	D0h	DFh	
		Studio	224	239	88%	94%	E0h	EFh	
		Reserved (see note 5 on page 15)	240	255	94%	100%	F0h	FFh	

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Lonestar DMX Channel Map

Lonestar Trifusion Frost Protocol

Channel	Function	Description	Decimal Low	Decimal High	Percent Low	Percent High	Hex Low	Hex High	Controller Default
1/2	Pan (Coarse/Fine)	See note 7 on page 16	0	65535	0%	100%	0000h	FFFFh	32767
3/4	Tilt (Coarse/Fine)	See note 7 on page 16	0	65535	0%	100%	0000h	FFFFh	32767
5	Color Mode	Full Speed Control							0
		Pure Mix	0	31	0%	12%	00h	1Fh	
		Cycle	32	47	13%	18%	20h	2Fh	
		Random	48	63	19%	25%	30h	3Fh	
		Pure Mix Fast	64	79	25%	31%	40h	4Fh	
		Cycle Fast	80	95	31%	37%	50h	5Fh	
		Random Fast	96	111	38%	44%	60h	6Fh	
		Reserved (see note 5 on page 15)	112	127	44%	50%	70h	7Fh	
		mSpeed Control							
		Pure Mix	128	159	50%	62%	80h	9Fh	
		Cycle	160	175	63%	69%	A0h	AFh	
		Random	176	191	69%	75%	B0h	BFh	
		Pure Mix Fast	192	207	75%	81%	C0h	CFh	
		Cycle Fast	208	223	82%	87%	D0h	DFh	
		Random Fast	224	239	88%	94%	E0h	EFh	
		Reserved (see note 5 on page 15)	240	255	94%	100%	F0h	FFh	
6 7 8	Cyan Magenta Yellow	Pure Mix							255
		Full Saturation to Open	0	255	0%	100%	00h	FFh	
		Cycle and Random. Scan speed controlled by Cyan channel.							
9	CTO	Slow to Fast	0	255	0%	100%	00h	FFh	255
		Full Saturation to Open	0	255	0%	100%	00h	FFh	

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Channel	Function	Description	Decimal Low	Decimal High	Percent Low	Percent High	Hex Low	Hex High	Controller Default	
10	Color Wheel Mode	Full Speed Control								48
		Indexed (see note 1 on page 15)	0	15	0%	6%	00h	0Fh		
		Forward Spin	16	31	6%	12%	10h	1Fh		
		Reverse Spin	32	47	13%	18%	20h	2Fh		
		Continuous (see note 1 on page 15)	48	63	19%	25%	30h	3Fh		
		Scan	64	79	25%	31%	40h	4Fh		
		Random	80	95	31%	37%	50h	5Fh		
		Reserved (see note 5 on page 15)	96	127	38%	50%	60h	7Fh		
		mSpeed Control								
		Indexed (see note 1 on page 15)	128	143	50%	56%	80h	8Fh		
		Forward Spin	144	159	56%	62%	90h	9Fh		
		Reverse Spin	160	175	63%	69%	A0h	AFh		
		Continuous (see note 1 on page 15)	176	191	69%	75%	B0h	BFh		
		Scan	192	207	75%	81%	C0h	CFh		
		Random	208	239	82%	94%	D0h	EFh		
		Reserved (see note 5 on page 15)	240	255	94%	100%	F0h	FFh		

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Channel	Function	Description	Decimal Low	Decimal High	Percent Low	Percent High	Hex Low	Hex High	Controller Default
11	Color Wheel	Indexed							0
		Open	0	10	0%	4%	00h	0Ah	
		Open/Red	11	21	4%	8%	0Bh	15h	
		Red	22	32	9%	13%	16h	20h	
		Red/Blue	33	43	13%	17%	21h	28h	
		Blue	44	54	17%	21%	2Ch	36h	
		Blue/Green	55	65	22%	25%	37h	41h	
		Green	66	76	26%	30%	42h	4Ch	
		Green/Yellow	77	87	30%	34%	4Dh	57h	
		Yellow	88	98	35%	38%	58h	62h	
		Yellow/Purple	99	109	39%	43%	63h	6Dh	
		Purple	110	120	43%	47%	6Eh	78h	
		Purple/TM-30	121	131	47%	51%	79h	83h	
		TM-30	132	142	52%	56%	84h	8Eh	
		TM-30/Lavender	143	153	56%	60%	8Fh	99h	
		Lavender	154	164	60%	64%	9Ah	A4h	
		Lavender/Half CTO	165	175	65%	69%	A5h	AFh	
		Half CTO	176	186	69%	73%	B0h	BAh	
		Half CTO/Full CTO	187	197	73%	77%	BBh	C5h	
		Full CTO	198	208	78%	82%	C6h	D0h	
		Full CTO/Dark Blue	209	219	82%	86%	D1h	DBh	
		Dark Blue	220	230	86%	90%	DCh	E6h	
		Dark Blue/Open	231	241	91%	95%	E7h	F1h	
		Open	242	255	95%	100%	F2h	FFh	
		Scan							
		Open/Red, Slow to Fast	0	21	0%	8%	00h	15h	
		Red/Blue, Slow to Fast	22	43	9%	17%	16h	28h	
		Blue/Green, Slow to Fast	44	65	17%	25%	2Ch	41h	
		Green/Yellow, Slow to Fast	66	87	26%	34%	42h	57h	
		Yellow/Purple, Slow to Fast	88	109	35%	43%	58h	6Dh	
		Purple/TM-30, Slow to Fast	110	131	43%	51%	6Eh	83h	
		TM-30/Lavender, Slow to Fast	132	153	52%	60%	84h	99h	
		Lavender/Half CTO, Slow to Fast	154	175	60%	69%	9Ah	AFh	
		Half CTO/Full CTO, Slow to Fast	176	197	69%	77%	B0h	C5h	
		Full CTO/Dark Blue, Slow to Fast	198	219	78%	86%	C6h	DBh	
		Dark Blue/Open, Slow to Fast	220	255	86%	100%	DCh	FFh	
		Spin and Random							
		Stop	0	0	0%	0%	00h	00h	
		Slow to Fast	1	255	0%	100%	01h	FFh	
		Continuous							
		Positioning from 0° to 359°	0	255	0%	100%	00h	FFh	

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Lonestar DMX Channel Map

Channel	Function	Description	Decimal Low	Decimal High	Percent Low	Percent High	Hex Low	Hex High	Controller Default	
12	Rotating Gobo Wheel Mode	Full Speed Control								0
		Indexed	0	15	0%	6%	00h	0Fh		
		Forward Spin	16	31	6%	12%	10h	1Fh		
		Reverse Spin	32	47	13%	18%	20h	2Fh		
		Scan	48	63	19%	25%	30h	3Fh		
		Random	64	79	25%	31%	40h	4Fh		
		Reserved (see note 5 on page 15)	80	127	31%	50%	50h	7Fh		
		mSpeed Control								
		Indexed	128	143	50%	56%	80h	8Fh		
		Forward Spin	144	159	56%	62%	90h	9Fh		
		Reverse Spin	160	175	63%	69%	A0h	AFh		
		Scan	176	191	69%	75%	B0h	BFh		
		Random	192	207	75%	81%	C0h	CFh		
		Reserved (see note 5 on page 15)	208	255	82%	100%	D0h	FFh		
13	Rotating Gobo Wheel (see list of gobos on page 17)	Indexed and Scan								0
		Open	0	15	0%	6%	00h	0Fh		
		Gobo 1 - Bar	16	31	6%	12%	10h	1Fh		
		Gobo 2 - Starvolver	32	47	13%	18%	20h	2Fh		
		Gobo 3 - Fracked	48	63	19%	25%	30h	3Fh		
		Gobo 4 - Broken Tunnel	64	79	25%	31%	40h	4Fh		
		Gobo 5 - Seashell	80	95	31%	37%	50h	5Fh		
		Gobo 6 - Glacial	96	111	38%	44%	60h	6Fh		
		Gobo 7 - Shower	112	127	44%	50%	70h	7Fh		
		Gobo 8 - Ice	128	143	50%	56%	80h	8Fh		
		Gobo 9 - Branch Out	144	159	56%	62%	90h	9Fh		
		Reserved (see note 5 on page 15)	160	255	63%	100%	A0h	FFh		
		Spin and Random								0
		Stop	0	3	0%	1%	00h	03h		
		Slow to Fast	4	255	2%	100%	04h	FFh		
14	Rotating Gobo Mode	Indexed	0	15	0%	6%	00h	0Fh	0	
		Forward Spin	16	31	6%	12%	10h	1Fh		
		Reverse Spin	32	47	13%	18%	20h	2Fh		
		Forward Animate	48	63	19%	25%	30h	3Fh		
		Reverse Animate	64	79	25%	31%	40h	4Fh		
		Reserved (see note 5 on page 15)	80	255	31%	100%	50h	FFh		
15/16	Rotating Gobo (Coarse/Fine)	Indexed								32767
		0° to 359°	0	65535	0%	100%	0000h	FFFFh		
		Spin and Animate								32767
		Stop	0	2047	0%	3%	0000h	07FFh		
Slow to Fast	2048	65535	3%	100%	0800h	FFFFh				
17	Blade 1 Angle A (see note 6 on page 15)	Removed to Inserted	0	255	0%	100%	00h	FFh	0	

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Lonestar DMX Channel Map

Channel	Function	Description	Decimal Low	Decimal High	Percent Low	Percent High	Hex Low	Hex High	Controller Default
18	Blade 1 Angle B (see note 6 on page 15)	Removed to Inserted	0	255	0%	100%	00h	FFh	0
19	Blade 2 Angle A (see note 6 on page 15)	Removed to Inserted	0	255	0%	100%	00h	FFh	0
20	Blade 2 Angle B (see note 6 on page 15)	Removed to Inserted	0	255	0%	100%	00h	FFh	0
21	Blade 3 Angle A (see note 6 on page 15)	Removed to Inserted	0	255	0%	100%	00h	FFh	0
22	Blade 3 Angle B (see note 6 on page 15)	Removed to Inserted	0	255	0%	100%	00h	FFh	0
23	Blade 4 Angle A (see note 6 on page 15)	Removed to Inserted	0	255	0%	100%	00h	FFh	0
24	Blade 4 Angle B (see note 6 on page 15)	Removed to Inserted	0	255	0%	100%	00h	FFh	0
25/26	Frame Rotation (Coarse/Fine)	-60° to 60°	0	65535	0%	100%	00h	FFFFh	32767
27	Animation Insertion	Removed to Inserted	0	255	0%	100%	00h	FFh	0
28	Animation	Forward Spin, Stop to Fast	0	63	0%	25%	00h	3Fh	0
		Reverse Spin, Stop to Fast	64	127	25%	50%	40h	7Fh	
		Forward Animate, Stop to Fast	128	191	50%	75%	80h	BFh	
		Reverse Animate, Stop to Fast	192	255	75%	100%	C0h	FFh	
29	Prism 1 Mode	Removed	0	15	0%	6%	00h	0Fh	0
		Continuous	16	31	6%	12%	10h	1Fh	
		Forward Spin	32	47	13%	18%	20h	2Fh	
		Reverse Spin	48	63	19%	25%	30h	3Fh	
		Forward Animate	64	79	25%	31%	40h	4Fh	
		Reverse Animate	80	95	31%	37%	50h	5Fh	
		Reserved (see note 5 on page 15)	96	255	38%	100%	60h	FFh	
30/31	Prism 1 Rotate (Coarse/Fine)	Continuous							32767
		0° to 359°	0	65535	0%	100%	0000h	FFFFh	
		Spin and Animate							0
		Stop	0	1027	0%	2%	0000h	0403h	
		Slow to Fast	1028	65535	2%	100%	0404h	FFFFh	
32	Reserved	Reserved for Future Use	0	255	0%	100%	00h	FFh	0
33/34	Diffusion Heavy (Coarse/Fine) (see note 4 on page 15)	Removed	0	0	0%	0%	0000h	0000h	0
		Hard to Soft Edge	1	65535	0%	100%	0001h	FFFFh	
35	Diffusion Light	Removed	0	0	0%	0%	00h	00h	0
		Hard to Soft Edge	1	255	0%	100%	01h	FFh	

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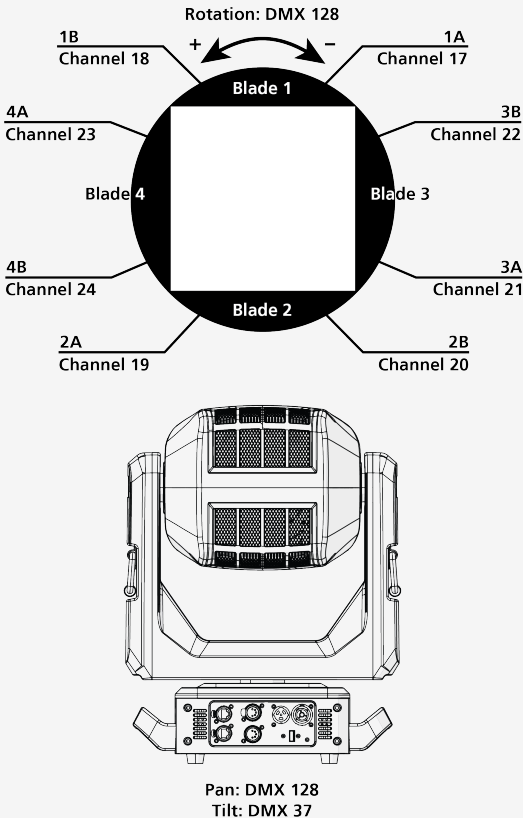
Lonestar DMX Channel Map

Channel	Function	Description	Decimal Low	Decimal High	Percent Low	Percent High	Hex Low	Hex High	Controller Default	
36	Diffusion Medium	Removed	0	0	0%	0%	00h	00h	0	
		Hard to Soft Edge	1	255	0%	100%	01h	FFh		
37/38	Focus (Coarse/Fine)	Focus	0	65535	0%	100%	00h	FFFFh	0	
39/40	Zoom (Coarse/Fine)	3.8° to 55°	0	65535	0%	100%	00h	FFFFh	0	
41	Iris	Closed to Open	0	255	0%	100%	00h	FFh	255	
42	Strobe Mode	Normal	0	15	0%	6%	00h	0Fh	0	
		Random	16	31	6%	12%	10h	1Fh		
		Synchronous Random (see note 2 on page 15)	32	47	13%	18%	20h	2Fh		
		Reserved (see note 5 on page 15)	48	255	19%	100%	30h	FFh		
43	Strobe	Closed	0	0	0%	0%	00h	00h	255	
		Slow to Fast	1	254	0%	100%	01h	FEh		
		Open	255	255	100%	100%	FFh	FFh		
44/45	Dimmer (Coarse/Fine)	Off to Full	0	65535	0%	100%	00h	FFFFh	0	
46	mSpeed	Not Enabled	0	3	0%	1%	00h	03h	0	
		Longest to Shortest	4	255	2%	100%	04h	FFh		
47	Control	No hold time requirement for the following functions (see note 8 on page 16):								0
		Idle	0	15	0%	6%	00h	0Fh		
		Pan and Tilt mSpeed Off	16	19	6%	7%	10h	13h		
		Reserved (see note 5 on page 15)	20	31	8%	12%	14h	1Fh		
		Hold time requirement for the following functions:								
		Display Off (hold 3 seconds)	32	47	13%	18%	20h	2Fh		
		Display On (hold 3 seconds)	48	63	19%	25%	30h	3Fh		
		Reserved (see note 5 on page 15)	64	79	25%	31%	40h	4Fh		
		Home All (hold 3 seconds)	80	95	31%	37%	50h	5Fh		
		Shutdown (hold 9 seconds)	96	111	38%	44%	60h	6Fh		
		Disable Pan/Tilt Motors (hold 3 seconds)	112	127	44%	50%	70h	7Fh		
		Dimming Mode - 2.4 kHz (hold 3 seconds)	128	143	50%	56%	80h	8Fh		
		Dimming Mode - 16 kHz (hold 3 seconds)	144	159	56%	62%	90h	9Fh		
		Reserved (see note 5 on page 15)	160	255	63%	100%	A0h	FFh		
48	Fan (see note 3 on page 15)	Idle	0	15	0%	6%	00h	0Fh	0	
		Set Continuous Speed - Slow to Fast	16	207	6%	81%	10h	CFh		
		Auto	208	223	82%	87%	D0h	DFh		
		Studio	224	239	88%	94%	E0h	EFh		
		Reserved (see note 5 on page 15)	240	255	94%	100%	F0h	FFh		

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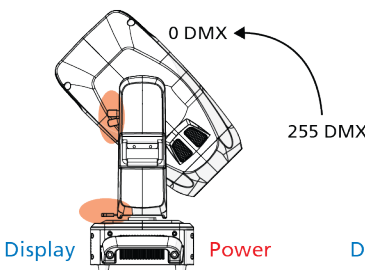
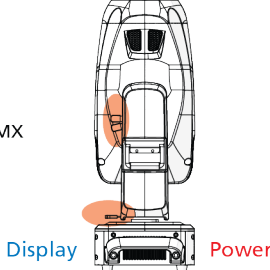
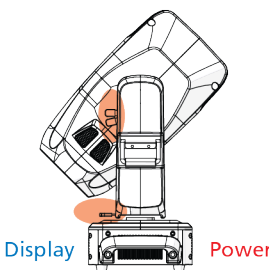
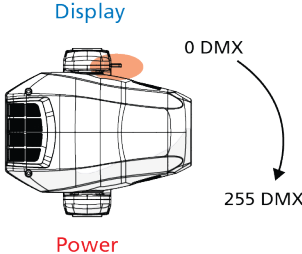
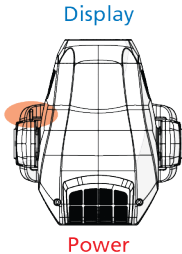
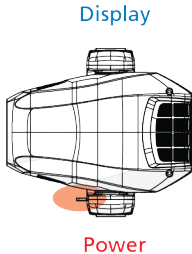
Lonestar DMX Channel Map

Notes

1	Continuous/Indexed mode should take the quickest path from 255–0 and 0–255. Continuous mode color wheel aperture centers are as follows: <table border="1"> <thead> <tr> <th>Color</th><th>Center of Color DMX Value</th></tr> </thead> <tbody> <tr><td>Open</td><td>0</td></tr> <tr><td>Red</td><td>26</td></tr> <tr><td>Blue</td><td>47</td></tr> <tr><td>Green</td><td>70</td></tr> <tr><td>Yellow</td><td>93</td></tr> <tr><td>Purple</td><td>116</td></tr> <tr><td>TM-30</td><td>138</td></tr> <tr><td>Lavender</td><td>161</td></tr> <tr><td>Half CTO</td><td>183</td></tr> <tr><td>Full CTO</td><td>206</td></tr> <tr><td>Dark Blue</td><td>229</td></tr> </tbody> </table>	Color	Center of Color DMX Value	Open	0	Red	26	Blue	47	Green	70	Yellow	93	Purple	116	TM-30	138	Lavender	161	Half CTO	183	Full CTO	206	Dark Blue	229
Color	Center of Color DMX Value																								
Open	0																								
Red	26																								
Blue	47																								
Green	70																								
Yellow	93																								
Purple	116																								
TM-30	138																								
Lavender	161																								
Half CTO	183																								
Full CTO	206																								
Dark Blue	229																								
2	Synchronous random strobes are synchronized across fixtures.																								
3	When the fixture is turned off it will retain the selected fan mode.																								
4	Standard Protocol: Prism 1 and Prism 2 can be inserted in the light path simultaneously. Trifusion Protocol: Diffusion filter is placed in Prism 2 holder. Prism 1 and Diffusion can be inserted in the light path at the same time.																								
5	Reserved ranges should function according to the controller default value.																								
6	Framing blades are arranged according to graphics below. For full-curtain framing, set blades 2A and 2B (channels 19 and 20) at or near 100%.  Pan: DMX 128 Tilt: DMX 37																								

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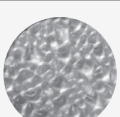
Lonestar DMX Channel Map

7	DMX Fixture Orientation (Tilt Movement Range: 251°; Pan Movement Range: 540°) (Pan and tilt lock locations are highlighted in orange.)		
			
	Tilt = 255 DMX Pan = 128 DMX	Tilt = 128 DMX Pan = 128 DMX	Tilt = 0 DMX Pan = 128 DMX
			
	Pan = 0 DMX Tilt = 37 DMX	Pan = 128 DMX Tilt = 37 DMX	Pan = 255 DMX Tilt = 37 DMX
	0 DMX 255 DMX		
8	Control channel functions with no hold time requirement must apply upon receipt.		
9	RDM Manufacturer ID: 0x4c52		
10	RDM Device ID: 0x05D0		

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Lonestar DMX Channel Map

Lonestar Rotating Gobos

Position	Name	Image
1	Bar	
2	Starvolver	
3	Fracked	
4	Broken Tunnel	
5	Seashell	
6	Glacial	
7	Shower	
8	Ice	
9	Branch Out	

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Lonestar DMX Channel Map

Revision History

Revision	Change	Release Date
F	Changes apply to both protocols: Added mSpeed Control to Color Mode (channel 5), Color Wheel Mode (channel 10), and Rotating Gobo Wheel 1 Mode (channel 12). Added fast modes to Color Mode (channel 5). Updated the Controller Default value for Color Wheel Mode (channel 10). Updated mSpeed values (channel 46). Added Pan and Tilt mSpeed Off to Control (channel 47), and identified the hold time requirements for the Control channel functions. Added notes 4 and 8 to the Notes section.	November 2023
E	Updated the name of the Trifusion protocol to "Trifusion Frost" to match the name in the fixture software. Updated the framing graphic and added full-curtain framing information to note 5.	June 2023
D	Corrected the Decimal High, Percent High, and Hex High values for channels 25 and 26 in both the Standard and Trifusion protocols.	May 2022
C	Added hold times to channel 47, Control, in both the Standard and Trifusion protocols.	February 2022
B	Corrected the center of color DMX values for red, blue, green, purple, TM-30, lavender, Half CTO, Full CTO, and dark blue.	October 2021
A	Initial release	August 2021